

Towards an Adaptive Approach for Broadcast Algorithms in MANETs

Raziel Carvajal-Gómez and Etienne Rivière

Université Catholique de Louvain

Louvain-la-Neuve, Belgium

name.surname@uclouvain.be

Abstract—Broadcast is a fundamental operation in Mobile Ad-Hoc Networks (MANETs). A large variety of broadcast protocols have been proposed to deal with network dynamics, maximize coverage and minimize the impact of collisions. These protocols follow either an unstructured approach based on the control of a flooding process or a more structured approach involving the creation of a dissemination overlay. The selection of the appropriate broadcast protocol for a deployed MANET is a difficult task. This paper presents the ongoing work towards a novel adaptive approach to broadcast in heterogeneous MANETs. This approach allows nodes to pick dynamically the most appropriate broadcast protocols for different regions in the communication area. We present preliminary results of an interoperability mechanism to broadcast in an scenario where more than one protocol is deployed. Our results show that the energy cost tends to reduce with such an approach, keeping a high coverage of nodes improving the network throughput.

I. INTRODUCTION

Mobile ad-hoc networks (MANETs) do not rely on fixed infrastructures for their deployment. Mobile nodes in such a network have wireless transceivers with a limited transmission range and forward packets to communicate with each other in a peer-to-peer fashion. Application scenarios such as vehicular networks [1], monitoring of zones in danger or the deployment of communication infrastructures in rural zones are suitable scenarios MANETs where broadcast is a fundamental operation.

Broadcast protocols for MANETs tackle the problem of selecting a set of forwarding nodes that retransmit received packets to reach every node in the network. Choosing a forwarding set is not an easy task, a small set reduces the overall energy consumption but may result in unreachable packets for other nodes. On the contrary, a bigger forwarding set increase the collisions and contentions in the network. Mobility is another factor, if a forwarding set is not updated according to the velocity of nodes, obsolete choices of this set may cause lost of packets. Additionally, mobility patterns may result in partitions and heterogeneous networks where the density of nodes varies within the area of communication.

We are interested in designing adaptive approaches for broadcast protocols in MANETs. In our context, adaptation refers to the ability of dynamically selecting and configuring the protocol use to broadcast. Notice that the deployment of a broadcast protocol may take place either in the entire system, or more interestingly, just in a part of it. To deal with the heterogeneous conditions of mobility and density,

we are working in the design of adaptive approaches that converge to hybrid systems: on one side, overlays emerge in highly dense regions where nodes remains mostly static and on the other side, controlled flooding protocols are deployed in sparse regions where nodes move more rapidly. In our vision, nodes discover (only with local information) the need of adaptation and perform a collective decision towards a switch to a different protocol. Our evaluation scenario consist of mobile devices hold by humans, we use the truncated-levy walk model as mobility pattern [2]. The mobile device follows the specifications of a transceiver frequently use in cellphones with IEEE 802.11 capabilities.

II. BACKGROUND

A basic approach to broadcast is *simple flooding* where every node retransmit an incoming packet just once, further copies of previously received packets are ignored. Simple flooding cause broadcast storms, term referring to a high rate of collisions and contentions in the network [3]. A vast amount of protocols have been proposed to deal with broadcast storms for finding an adequate trade-off between reachability and usage of nodes resources (mainly energy consumption). P Ruiz and P Bouvry [4] classify broadcast protocols in two groups. Protocols of the first group build and maintain an overlay via a periodic exchange of control messages that announce the existence of nodes. Overlay-based approaches may form trees [5], clusters or connected-dominating sets (CDS), where every node is either part of a CDS or adjacent to at least one member of this set [6]. In other words, CDS are forwarding sets and every other node out of it acts as receiver. The second group of broadcast protocols do not rely on overlays. Instead, the retransmission of packets take place following a stochastic approach [7] or when predefined thresholds reach their maximum value (ex., received packets).

We compare our hybrid approach to the most representative protocols of each category presented in [4]. Our work is inspired from an evaluation study of broadcast protocols for MANETs [8] driven by a plethora of configurations of mobility and density. This study reveals that overlay-based approaches perform better in a dense network where nodes remain mostly static. On the contrary, in a sparse network where nodes move faster non overlay-based approaches show better results.

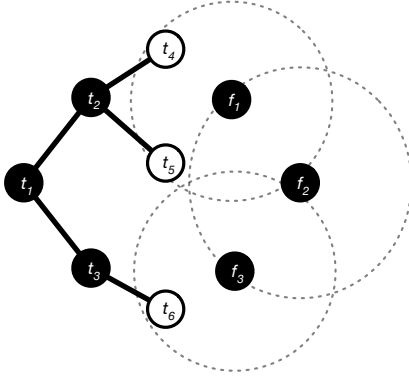


Fig. 1. Hybrid approach where blind flooding and a tree-based broadcast approach runs in a MANET.

III. INTEROPERABILITY IN BROADCAST PROTOCOLS

We assume that a set $\mathbb{P}$ of broadcast protocols is in execution in a deployed MANET of $\mathbb{N}$ nodes where each node $n_i \in \mathbb{N}$ runs only one protocol $p_j \in \mathbb{P}$ at a time. Our aim is letting a node n_i chose in an autonomous way a certain protocol p_j based on the local information available at n_i . Such information may be, and it is not limited to, remaining battery, signal strength of received packets or node's velocity. But we omit the case where n_i is running more than one protocol concurrently.

When $\mathbb{P}$ contains only one protocol the reachability of messages depends on the performance of such approach. Notice that in our hybrid approach we do not intent to alter the internals of any protocol in $\mathbb{P}$. On the contrarily, when the cardinality of $\mathbb{P}$ is $|\mathbb{P}| > 1$ packets may be lost. Figure 1 depicts a situation where a set of nodes $\mathbb{T} = \{t_i \mid i \in \{1, \dots, 6\}\}$ run a topology-based protocol while nodes in $\mathbb{F} = \{f_j \mid j \in \{1, \dots, 3\}\}$ run simple flooding. Broadcast sessions initiated at t_1 form a tree such that a forwarding set composed of t_1, t_2, t_3 guarantees the message reception only for nodes at $\mathbb{T}$. Even if the leafs of this tree t_4, t_5, t_6 are within the range of some nodes in $\mathbb{F}$, their role as pure receivers in the tree avoids any node in $\mathbb{F}$ hearing from messages.

A. Detection of border nodes

Algorithm 1 is a simple mechanism we propose to deal with interoperability issues between nodes running different protocols. We intent to increase the set of forwarding nodes from only one running protocol with a set we called *border nodes*. Intuition says that the bigger the size of this set the higher the reachability of nodes among broadcast sessions. Border nodes are within the transmission range of other nodes running a different protocol. In figure 1 nodes t_4, t_5, t_6 are candidates of border nodes. We consider that *broadcast messages* are vectors $(\iota, \tau, \rho, \varrho)$ with four entries where: ι is a unique identifier, τ defines either a control or broadcast

Algorithm 1: Interoperability mechanism to deal with more than one running protocol in a MANET.

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1 variables
2 node_id : string // node identifier
3 running_alg : string // running algorithm
4 timeout: double // maximum waiting time to start the
  procedure of border detection
5 am_i_border : boolean // says wether the local node
  is a border node or not
6 known_algs : set // previously known foreign
  protocols
7 timers : dictionary // timers of border detection
  procedure
8 known_nodes : dictionary // previously known nodes
  from a foreign protocol

9 function receive(msg)
10 if msg.sender_alg = running_alg or am_i_border then
11   broadcast(msg)
12 else
13   switch msg do
14     case ControlMsg do
15       known_nodes[msg.sender_alg] ← msg.sender_id
16       if msg.sender_alg ∉ known_algs then
17         known_algs ← known_algs ∪ msg.sender_alg
18         delay ← new Random(0, timeout)
19         timers[msg.sender_alg] ← new Timer(delay)
20         when delay expires
21           b ← new BorderMsg()
22           b.chosen_node ← known_nodes[msg.sender_alg].pickOne()
23           b.fwd_no ← MAX_FWD_NO
24           b.sender_id ← node_id
25           b.sender_alg ← running_alg
26           b.foster_by ← msg.sender_alg
27           forward(b)
28     case BorderMsg do
29       msg.fwd_no ← msg.fwd_no - 1
30       if msg.fwd_no ≠ 0 then
31         forward(msg)
32       if running_alg = msg.foster_by and msg.foster_by ∉
33         known_algs then
34         known_algs ← known_algs ∪ msg.foster_by
35         if node_id = msg.chosen_node then
36           am_i_border ← True
37       if running_alg = msg.sender_alg and msg.sender_alg ∉
38         known_algs then
39         known_algs ← known_algs ∪ msg.sender_alg
40         if timers[msg.sender_alg] ≠ ∅ then
41           cancelTimer(timers[msg.sender_alg])

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message, ρ refers to the running protocol at the sender and ϱ is the message payload. *Control messages* on the other hand are beacons required by some protocols to operate.

Our interoperability mechanism works as follows. When nodes receive a message msg that corresponds to the running protocol of the sender or the actual receiver was previously labeled as border node (line 10), msg is broadcast; notice that the function *broadcast* (line 11) hides the protocol's implementation. In case the running protocols of sender and receiver differs, we distinguish between two cases (line 13):

- **Control message reception.** This case initiates the elec-

tion of border nodes. We keep track of senders from a foreign algorithm with the set *known_nodes* (line 8), potential candidates of border nodes are within this set. If receivers previously hear a beacon from a sender running a known foreign protocol, *msg* is ignored. Otherwise, we keep a reference of the known protocol (line 17) and a timer initiates (line 19). When the timer expires (line 20), we create a new border message *b* (line 21) with the following fields:

- *border node* (line 22): for the moment this is a random choice; alternatively, we may piggyback senders' remaining battery in control messages to select a border node with the highest value of energy
- *traveled hops* (line 23): number of times *b* will be forwarded
- *sender identifier* (line 24)
- *sender's running protocol* (line 25)
- *foreign protocol* (line 26): nodes running this protocol foster the border election procedure

We now explore the second case when nodes receive border messages (line 28).

- **Border message reception.** Receivers decrease by one the number of traveled hops (line: 29). In case this counter has remaining hops, *msg* is forward (lines: 30, 31). Notice that the function *forward* simply performs a retransmission within the communication range of the receiver. Given that all packets have in a unique identifier further copies of previously known packets are ignored. Furthermore, only one out of two possible situations take place. Either the receiver is a potential candidate of border node (line: 32) or it belongs to those nodes who foster the selection procedure (line 36). The latter situation serves to cancel ongoing timers that triggered the border selection procedure by other senders (line: 39), that is, nodes running a foreign algorithm. When the former situation occurs, the receiver is labeled as border node only if its identifier corresponds to the one in the message (line 35).

B. Evaluation scenario

We simulate a situation where a crowd of humans carry mobile devices with WiFi capabilities (cellphones, tablets, etc) and the area of communication has two regions, dense and sparse. Regions contain 400 nodes each and approximately the same surface, the dimensions of the whole area are 120 meters of height and width. The aim of having such an arrangement is measuring the impact of increasing the forwarding set of one protocol with border nodes found with Algorithm 1. Nodes are spread uniformly in both regions and their initial position is chosen in a random way. The number of nodes neighbors within the sparse region varies between 1 to 10 while those nodes in the dense region may have up to 40 neighbors.

Static scenario. The evaluation of Algorithm 1 takes place in a scenario where nodes do not move. In our future evaluation we will consider a dynamic scenario where nodes move at a

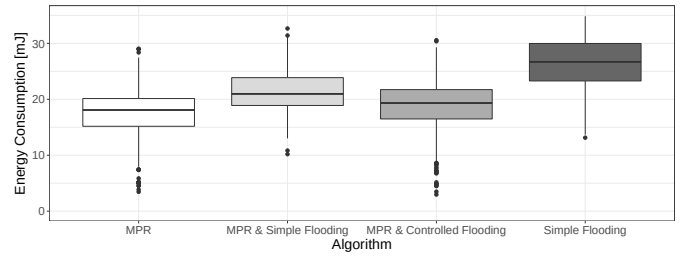


Fig. 2. Distribution of the energy consumed by 800 nodes over 200 broadcast sessions. The mean is indicated with a horizontal line on each box.

speed of 0.5 m/s within the dense region and the speed of those at the sparse region will be 2.0 m/s. These rates of velocity are based on an estimation of the preferred walking speed of humans [9].

Broadcast protocols. Running protocols on each node are chosen in advance and nodes keep using this protocol until the simulation concludes. Nodes in the sparse area use controlled flooding. When a node receives a broadcast message for the first time, it starts a timer with a duration chosen randomly between the interval $[0, \dots, max]$. In our experiments the threshold *max* is set to 10 ms. When the number of received copies per message reaches a maximum number of receptions before the timer expires, the message is not forwarded. Otherwise, the retransmission take place. This approach was first propose in [3] to deal with broadcast storms. Nodes in the dense area use an approximation of a connected-dominating [10].

Broadcast sessions. A total of 200 broadcast sessions are performed with a frequency of one second. The source of each sessions is chosen randomly among all nodes. The length of each message is 32 bytes.

Simulator. We deploy our simulation with OMNET++4.6 [11] and the plug-in INET3.3 to model MANETs. This framework provides a full implementation of the IEEE 802.11g standard, we use the standard energy consumption model of this simulator.

Transceiver settings. Nodes have a transmission range of 10 m. We model message transmission using the specifications of a commercial chip found in modern mobile devices, that is, the chip Broadcom BCM4329 [12]; this chip supports Wi-Fi 802.11g on the 2.4 GHz frequency band. Its energy consumption per operation modes is as follows: sleep (1mW), receive (10mW) and transmit (100mW).

C. Preliminary Results

We deploy the static scenario described in section III-B to measure: energy consumption, throughput of broadcast messages and reachability. We perform 4 independent executions: simple flooding, the overlay-based approach MPR (multi-point relaying technique [10]) and two hybrid configurations. MPR runs in the dense area of both hybrid configurations whereas simple flooding and controlled flooding run in the sparse region of each configuration, respectively.

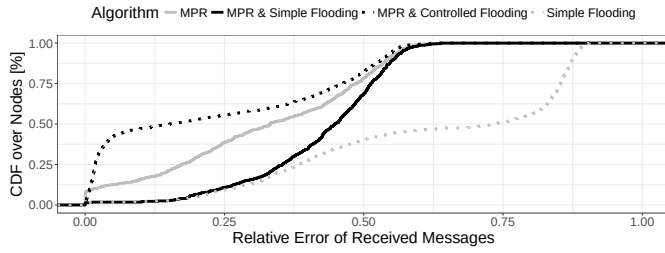


Fig. 3. CDF of relative error for received messages.

Energy consumption. Figure 2 reveals that MPR reduces the required energy to disseminate 200 broadcast sessions in a static wireless topology. We observe that the forwarding set built per session contains up to 30% of the nodes in the network while approximately 70% of nodes act as receivers. On the other hand, in a simple flooding approach where every node retransmits at least once leads in the lowest bound of performance for this metric. The average consumed energy in our two hybrid approaches (21 mJ and 19.5 mJ, respectively) remains closer to the one observed in MPR (17.9 mJ).

Relative Error (RE) of messages throughput. For this metric, we obtain the wireless topology formed at each broadcast session to get the number of neighbors per node. With this ground truth and the number of received broadcast messages at each node, we show the distribution of the RE of received messages in form of a CDF in figure 3. We observe that the throughput of receptions increases in one hybrid approach (dotted line in black) where controlled flooding is running in 50% of the nodes at the sparse region. The periodic exchange of beacons in MPR, which approximates surroundings, reduces the throughput in a deployed MANET (solid line in gray). As expected, simple flooding shows the lowest performance (dotted line in gray).

Reachability of broadcast messages. Figure 4 shows another perspective of reached nodes over all broadcast sessions. Once again, we observe that a flooding approach in an sparse region increases the reachability of messages (solid line in black). Up to 94% of nodes receive all broadcast messages and 3% of nodes miss up to 1% of messages. Notice that even in a static scenario, we can not guarantee completely reachability due to the presence of collisions.

IV. CONCLUSION AND FUTURE WORK

Selecting which is the appropriate broadcast algorithm for a given MANET deployment scenario is a daunting task. The performance of a given approach depends on its deployment context. This makes hard to find the right balance between reliability and efficiency. We present preliminary results of an hybrid approach for broadcast protocols in MANETs where an overlay is deployed in dense regions of the network and controlled flooding techniques run in sparse regions.

As future work, we will focus in the specifications of adaptation rules for letting nodes decide which broadcast protocol fits better in the network. This decision may be collective or

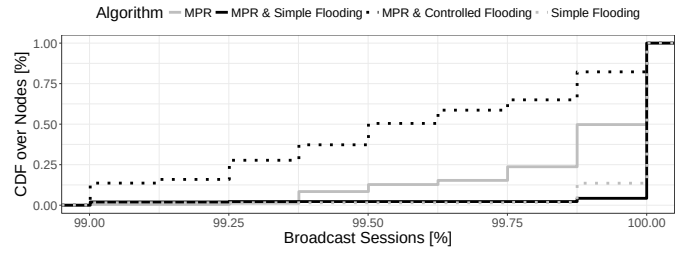


Fig. 4. Reachability of broadcast messages.

individual but completely decentralized. We rely on the local meta data available in nodes such as: the signal strength of received packets, remaining energy or rate of received packets. Mainly, nodes velocity and density are two transversal aspects to take into consideration as well.

ACKNOWLEDGMENT

The research leading to these results has received funding from CHIST-ERA under project DIONASYS, from the Swiss National Science Foundation (SNSF) under grant 155249, and from the French National Research Agency (ANR) under grant ANR-14-CHR2-0004.

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